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CagA of *Helicobacter pylori* interacts with and inhibits the serine-threonine kinase PRK2

Jyoti Prasad Mishra[1], **David Cohen**[2], **Andrea Zamperone**[2], **Dragana Nesic**[3], **Anne Muesch**[2], and **Markus Stein**[1,*]

[1]Department of Health Sciences, School of Arts and Sciences, Albany College of Pharmacy and Health Sciences, Albany, NY, USA.

[2]Albert Einstein College of Medicine, Bronx, NY USA.

[3]Laboratory of Structural Microbiology, The Rockefeller University, New York, NY, USA.

Summary

CagA is a multifunctional toxin of *Helicobacter pylori* that is secreted into host epithelial cells by a type IV secretion system. Following host cell translocation, CagA interferes with various host–cell signalling pathways. Most notably this toxin is involved in the disruption of apical–basolateral cell polarity and cell adhesion, as well as in the induction of cell proliferation, migration and cell morphological changes. These are processes that also play an important role in epithelial-to-mesenchymal transition and cancer cell invasion. In fact, CagA is considered as the only known bacterial oncoprotein. The cellular effects are triggered by a variety of CagA activities including the inhibition of serine–threonine kinase Par1b/MARK2 and the activation of tyrosine phosphatase SHP-2. Additionally, CagA was described to affect the activity of Src family kinases and C-terminal Src kinase (Csk) suggesting that interference with multiple cellular kinase- and phosphatase-associated signalling pathways is a major function of CagA. Here, we describe the effect of CagA on protein kinase C-related kinase 2 (PRK2), which acts downstream of Rho GTPases and is known to affect cytoskeletal rearrangements and cell polarity. CagA interacts with PRK2 and inhibits its kinase activity. Because PRK2 has been linked to cytoskeletal rearrangements and establishment of cell polarity, we suggest that CagA may hijack PRK2 to further manipulate cancer-related signalling pathways.

Introduction

In 2005, the Australian scientists Barry and Marschall received the Nobel Prize for discovering the association between gastric colonization with *Helicobacter pylori* and peptic ulcer disease, which until then was thought to be a stress-related event (Marshall and Warren, 1984b; Marshall *et al*., 1984a). Today, it is broadly accepted that *H. pylori*, which colonizes approximately two-thirds of the world human population, causes not only peptic and duodenal ulcers but also mucosa-associated lymphoid tissue lymphoma and gastric cancer (el Omar *et al*., 1995; El-Omar *et al*., 1997; Ogura *et al*., 2000; Han *et al*., 2002; Herrera and Parsonnet, 2009; Konturek *et al*., 2009; Malfertheiner, 2011). Although *H.*

*For correspondence. markus.stein@acphs.edu; Tel. (+518) 694 7174; Fax (+518) 694 7382..

pylori expresses various virulence proteins, the presence of the *cag*–pathogenicity island (*cag*–PAI) appears to be the major factor responsible for the development of duodenal and gastric ulcers and gastric cancer (Covacci *et al*., 1993; 1997; Censini *et al*., 1996; Akopyants *et al*., 1998; Ohnishi *et al*., 2008; Yamaoka, 2010). Encoded in the *cag*–PAI are a type IV secretion system (T4SS) and its substrate, the CagA cytotoxin, which is a unique protein with no known homologue (Crabtree *et al*., 1991; Peek *et al*., 1995; Censini *et al*., 1996). Studies have shown that after colonization of the stomach, the T4SS mediates the translocation of CagA into the host epithelial cells, where it associates with the cytoplasmic membrane (Segal *et al*., 1999; Asahi *et al*., 2000; Backert *et al*., 2000; Odenbreit *et al*., 2000; Stein *et al*., 2000).

Although the exact mechanism of CagA translocation remains elusive, it was shown that the process is triggered via the interaction between the pilus adhesin of the T4SS, CagL and the α5β1 integrin receptor, and also requires the association between CagA and phosphatidylserine localized at the outer leaflet of the host cytoplasmic membrane (Kwok *et al*., 2007; Jimenez-Soto *et al*., 2009; Murata-Kamiya *et al*., 2010; Conradi *et al*., 2012; Kaplan-Turkoz *et al*., 2012). After internalization into the host cell, CagA associates with eukaryotic proteins leading to the interference with various host-signalling pathways. Importantly, CagA causes the disruption of epithelial cell-to-cell adhesions, and the formation of pseudopodia and stress fibres resulting in elongations referred to as the hummingbird phenotype (Segal *et al*., 1999; Amieva *et al*., 2003; Bagnoli *et al*., 2005; Stein *et al*., 2013; Hatakeyama, 2014). Specifically, CagA disrupts tight junctions via inhibition of the serine–threonine kinase Par1b-MARK2 (Saadat *et al*., 2007; Zeaiter *et al*., 2008) and adherence junctions via the direct interaction with E-cadherin (Murata-Kamiya *et al*., 2007; Kurashima *et al*., 2008; Oliveira *et al*., 2009). These CagA functions depend on the C-terminal dimerization motifs (TM motifs), which directly bind to the kinase active site of Par1b thereby inhibiting kinase activity (Nesic *et al*., 2014). Furthermore, CagA affects mitogenic events and the integrity of focal adhesions through binding of the EPIYA motifs of CagA to the SH2 domain of the tyrosine phosphatase SHP-2 (Higashi *et al*., 2002). After binding to CagA, SHP-2 induces cell proliferation via the activation of the extracellular signal-regulated kinase/mitogen-activated protein kinase signalling pathway, morphological changes of the host cell known as ‘hummingbird phenotype’, cell motility and detachment of cells from cultured dishes. The latter effects are mostly influenced by SHP-2-dependent dephosphorylation of focal adhesion kinase, which leads to a decrease of focal adhesion sites (Tsutsumi *et al*., 2006). CagA/SHP-2 interaction requires phosphorylation of the EPIYA motifs on the tyrosine residues by Src family kinases and c-Abl (Higashi *et al*., 2002; Stein *et al*., 2002; Poppe *et al*., 2007). Different strains of *H. pylori* can contain different numbers of EPIYA and TM motifs as both motifs are located within a carboxy-terminal repeat region of CagA (Yamaoka and Graham, 2001). Interestingly, an increased number of motifs seem to correlate with an enhanced ability of CagA to interfere with host signalling (Naito *et al*., 2006).

Additionally, CagA affects other signalling molecules including Src family kinases (Stein *et al*., 2002; Selbach *et al*., 2003; Tegtmeyer and Backert, 2011), C-terminal Src kinase (Csk)

(Tsutsumi *et al*., 2003), hepatocyte growth factor receptor/c-Met (Churin *et al*., 2003) and the adaptor molecules Grb2 (Mimuro *et al*., 2002) and Crk (Suzuki *et al*., 2005).

*P*rotein kinase C-*r*elated *k*inase 2 (PRK2) is an isoform of a family of serine–threonine kinases that has been isolated first from a human hippocampal cDNA library in 1994 (Mukai and Ono, 1994). Shortly after, several isoforms of the kinase were identified in human and animal tissues and different names were assigned to the isoforms. The three isoforms identified in humans are PRK1 (PKNα, PAK-1), PRK2 (PKNƔ, PAK-2) and PRK3 (PKNβ). Although PRK1 and PRK2 were found ubiquitously expressed in most human organs, PRK3 expression was mainly restricted to various cancer cells, skeletal muscle, heart and liver tissues (Palmer *et al*., 1995). Structural features of PRK include three amino-terminal *a*ntiparallel-*c*oiled *c*oil (ACC) domains with binding affinity for small rho family GTPases (ACC1 and ACC2) (Maesaki *et al*., 1999), which are followed by a C2-like domain with similarity to the calcium-dependent C2 phospholipid binding domain of PKC_{ε} and PKC_{η} (Vincent and Settleman, 1997; Flynn *et al*., 1998; Mukai, 2003). The latter domain is presumed to have autoregulatory functions or, alternatively, may serve as a protein–protein interaction motif (Mukai, 2003). In PRK2 a proline-rich region, which is duplicated in PRK3 but absent in PRK1, is located in the linker between the C2-like domain and the carboxy-terminal kinase domain (Mukai, 2003). The kinase domain is highly homologous between various PRK isoforms and also protein kinase C (PKC). Indeed, all these kinases phosphorylate similar substrates *in vitro*. Thus, the specificity for each kinase activity likely results from differences in tissue-specific expression levels, subcellular distribution, regulation of kinase activity and exposure to different substrates. Substrates that have been described for PRK2 include cortactin (Bourguignon *et al*., 2007), Nck4 (Quilliam *et al*., 1996), Grb4 (Braverman and Quilliam, 1999) and Fyn (Calautti *et al*., 2002). Specifically, PRK2 has been suggested to regulate apical junction and cadherin-dependent cell-to-cell contact formation, thus contributing to increased cell adhesion (Calautti *et al*., 2002; Wallace *et al*., 2011). In agreement, only one PRK homologue exists in *Drosophila*, which is thought to regulate epithelial morphogenesis downstream of Rho through regulation of cell-to-cell adhesions and actomyosin-dependent cell shape changes (Lu and Settleman, 1999). However, an involvement of PRK2 in cell cycle control (Schmidt *et al*., 2007), and cell migration and cell invasion (Leenders *et al*., 2004; Lachmann *et al*., 2011), has also been demonstrated.

Phosphoinositide-dependent kinase 1 (PDK1) was shown to activate PRK1 and also PRK2 activities via phosphorylation of the activation loop (Belham *et al*., 1999; Dettori *et al*., 2009). The amino-terminal region of PRK2 strongly inhibits PRK2 kinase activity by mediating PRK2–PRK2 oligomerization, which blocks PDK1 binding to the C-terminal PDK1 interacting fragment (PIF) of PRK2 thereby preventing PRK2 activation (Bauer *et al*., 2012). This mechanism is different from the regulation of other PKC-type kinases that are typically activated by PDK1 via intramolecular binding of the amino-termini. PRK2 activation requires binding of Rho-GTP to the ACC1 domain, which then causes the release of the PIF fragment allowing PDK1 binding and phosphorylation of the PRK2 activation loop (Bauer *et al*., 2012). After further phosphorylation events, PDK1 dissociates and the active kinase structure is stabilized by intramolecular interaction between the PIF fragment and the catalytic domain. These events likely trigger the dissociation of the PRK2 multimers and release the active PRK2 monomers (Bauer *et al*., 2012).

In the present work we show that CagA interacts with PRK2 and inhibits PRK2 activity. Although the cellular function of PRK2 inhibition by CagA remains unknown, we suggest that PRK2 inhibition by CagA may corroborate with established CagA functions, thus contributing to cytoskeletal rearrangements, disruption of cellular polarity and reduction of overall cell-to-cell adhesion. The interference of CagA with cell adhesion and growth factor-like responses is reminiscent of the process of epithelial-tomesenchymal transition (EMT), a complex growth factor-induced phenotype that is defined by disruption of cell-to-cell adhesion, reorganization of the cytoskeleton into pseudopodial structures and increase of cell motility (Polyak and Weinberg, 2009). Because EMT has been identified to play an important role in cancer development, CagA may affect similar pathways to cause malignancy (Bagnoli *et al*., 2005; Saito *et al*., 2010; Yin *et al*., 2010; Stein *et al*., 2013). The inhibition of PRK2 by CagA may play an important role in gastric disease by supporting these CagA-mediated oncogenic activities.

Results

CagA recruits PRK2 to AGS cell membrane fractions

Many crucial cellular signalling pathways involve one or more protein phosphorylation and/or dephosphorylation steps. To gain new insights into CagA-dependent host-signalling events, we purified phosphoproteins from AGS cell membrane fractions following infection with either the G27 wild-type strain of *H. pylori* or, alternatively, with an isogenic *cag*A deletion mutant (Δ*cag*A). The phosphoproteins were then analysed by iTRAQ mass spectrometry to detect relative differences in membrane-associated phosphoprotein profiles. The two phosphoproteins that appeared most enriched in membranes in the presence of CagA were the serine/threonine kinases PRK2 and Par1b/MARK2. Because a role for Par1b/MARK2 in CagA signalling had been previously established, we focused our attention on PRK2, in spite of the fact that the *P*-values of the iTRAQ analysis for PRK2 quantitative differences between CagA positive and negative cells were statistically not significant.

To test the significance of the PRK2 iTRAQ results, we infected AGS cells with G27 wild type or alternatively with two isogenic mutants: a *cag*A mutant (Δ*cag*A) and a mutant that did not contain the first two repeats and also was phosphorylation deficient (Δ*rep*EPISA) (Fig. 1 PY panel). After 4 h of infection, the cells were fractionated into cytosol and membrane and the proteins in each fraction were analysed by immunoblotting using various antibodies. In non-infected cells or cells infected with Δ*cag*A, PRK2 localized exclusively in the cytoplasmic fraction. However, when cells were infected with the *H. pylori* wild-type or phosphorylation-resistant Δ*rep*EPISA, the majority of PRK2 was redistributed to the membrane fraction (Fig. 1). Under similar conditions, CagA did not cause the redistribution of PRK1 kinase to the membrane fraction. PRK1 is another isoform of the PKN kinase family, which shows a high homology with PRK2 in the kinase domain but not in the regulatory domain. This finding indicates that recruitment of PRK2 by CagA was isoform specific (Fig. 1). Use of phospho-Prk1/2 antibody confirmed these findings. Only the upper band of the duplet corresponding to PRK2 showed redistribution to the membrane fraction, whereas the lower band corresponding to PRK1 remained in the cytosol fraction (Fig. 1). The data obtained with the phospho-specific PRK1/2 antibody also suggested that CagA did not increase PRK2 kinase activity. This conclusion was made, as phospho-specific PRK1/2 antibody recognizes phosphorylation of the PRK2 kinase activation loop, which is required for kinase activation. We did not observe an obvious change in activation loop phosphorylation but rather redistribution of phospho-PRK2 from the cytosol to the membrane (Fig. 1). Anti-SHP-2 antibody was used as a gel loading control.

Similar results were obtained when infection experiments were analysed by confocal (Fig. 2A) and fluores-cent microscopy (Fig. 2B). Cells infected with G27 for 4 h caused accumulation of PRK2 and phosphoPRK1/2 in proximity to the attaching bacteria (Fig. 2A and B). When cells were infected with an isogenic *cag*A mutant no accumulation of PRK2 beside the bacteria was observed (Fig. 2A). Simultaneous staining of AGS cells infected with wild-type bacteria using anti-phospho-PRK1/2 and anti-Par1b/MARK2 antibodies furthermore demonstrated complete co-localization of both kinases (Fig. 2B).

Together, these results indicate that CagA translocation into host cells is followed by specific recruitment of PRK2, but not of PRK1, from the cytosol to the membrane where it localizes beneath the attaching bacteria. PRK2 recruitment was independent of the phosphorylation status of CagA and similar to results previously described for Par1b/MARK2.

CagA recruits PRK2 and Par1b/MARK2 independently from each other. The previous experiments showed that CagA causes the redistribution of PRK2 to the AGS cell membrane fraction independent of CagA tyrosine phosphorylation. Because this redistribution pattern was similar to what we previously observed for Par1b/MARK2 (Zeaiter *et al*., 2008) (Fig. 2B), we next investigated whether Par1b/MARK2 recruitment was required prior to PRK2 recruitment by CagA or, vice versa, whether Par1b recruitment depended on prior interaction between PRK2 and CagA.

A strain was generated that expressed a *cag*A gene carrying point mutations in the crucial Par1b/MARK2 binding motif (ΔAxA-FLP). The Par1b/MARK2 binding motif of CagA is located C-terminal of the EPIYA motif and has previously been designated the multimerization motif (TM motif). When using the AGS cell infection model, the ΔAxA-FLP strain still translocated CagA (CagA-FLP) into host cells, where it was phosphorylated on tyrosine within the EPIYA motifs (data not shown). As expected, the isogenic strain expressing wild-type CagA (ΔAxA) mediated recruitment of Par1b/MARK2 to the membrane, whereas the Par1b binding-deficient strains (ΔAxA-FLP) did not (Fig. 3A). However, PRK2 was recruited to the membrane fraction of AGS cells that were infected with either strain ΔAxA or the ΔAxA-FLP mutant (Fig. 3A). Because the strains ΔAxA or the ΔAxA-FLP had been constructed using CagA of strain 26695, this strain was also included in the experiment and, as expected, behaved similar to the G27 wild-type strain.

Vice versa, to test a possible dependence of Par1b/MARK2 recruitment on PRK2 we used silencing RNA to suppress PRK2 expression in AGS cells (Fig. 3B). Cells were then infected with G27 wild-type or the isogenic *cag*A mutant followed by fractionation into cytosol and membrane. The fractions were analysed by immunoblotting using anti-CagA, anti-Par1b/MARK2 and anti-pPRK1/2 antibodies. Although PRK2 expression was greatly reduced in the cytosol as well as in the membrane fractions of wild-type infected cells, no effect was observed on CagA-mediated Par1b/MARK2 recruitment.

We also analysed the effect of PRK2 knock-down on the morphology of AGS cells or AGS cells that were infected with strain G27 using phase contrast microscopy, but did not notice any significant differences compared with infected or uninfected cells that were treated with control silencing RNA (data not shown). This suggests that PRK2 silencing did not have a major effect on the hummingbird phenotype, which as previously demonstrated (Higashi *et al*., 2002) depends on CagA interaction with SHP-2 and Par-1b/MARK2 (Yamahashi and Hatakeyama, 2013).

Together, our findings suggest that both kinases, Par1b/MARK2 and PRK2, are recruited to the bacterial attachment site independently from each other. Although the TM multimerization motif is required for Par1b/MARK2 binding, recruitment of PRK2 was independent of the TM motif, suggesting a different CagA domain as binding site for PRK2.

CagA directly interacts with PRK2—Although our experiments suggested that CagA recruits PRK2 to the host cytoplasmic membranes, where it is localized underneath the attaching bacteria, it was not clear whether this redistribution was mediated by a direct interaction between CagA and PRK2 or rather was mediated by additional bacterial or host cell proteins. To investigate a possible direct interaction between CagA and PRK2, we attempted immunoprecipitation studies using purified PRK2 and CagA that was partially purified from strain ΔAxA or alternatively from strain ΔAxA-FLP using hydroxylapatite resin. Although this method did not allow complete purity of CagA, it greatly enriched CagA and strongly reduced the presence of other bacterial proteins.

Figure 4 shows that anti-PRK2 antibody coimmunoprecipitated CagA and also the Par1b binding-deficient CagA-FLP (expressed by strain ΔAxA-FLP) when PRK2 was added to the reaction, but not in the absence of PRK2. Therefore, co-immunoprecipitation of CagA occurred via binding to PRK2, but not by unspecific binding of CagA to the resin or by cross-reactivity with PRK2 antibody. Because no other cellular proteins were present in the reaction, these results suggested that CagA recruits PRK2 independently of other host factors.

CagA inhibits PRK2 kinase activity—Because CagA appeared to directly interact with PRK2, the next question was whether CagA would affect the kinase activity of PRK2. We used partially purified CagA and active recombinant PRK2 to investigate the effect of CagA on PRK2 kinase activity using an *in vitro* kinase assay. Figure 5A shows that the presence of partially purified CagA significantly inhibited PRK2 kinase activity. In contrast, bovine serum albumin (BSA) did not affect PRK2 kinase activity. To demonstrate that the inhibitory effect was really due to CagA and not due to other *Helicobacter* proteins that were co-purified with CagA by the hydroxylapatite resin, we also used the same method that was used for partial CagA purification from wild-type bacteria to mock purify CagA from the isogenic *cag*A mutant. However, the hydroxylapatite binding proteins that were purified from the cagA mutant did not inhibit PRK2 kinase activity (Fig. 5A). The binding of the TM motif of CagA to the substrate-binding site of the Par1b leads to inhibition of Par1b kinase activity, thereby disrupting cellular polarity (Nesic *et al*., 2010). To confirm our previous finding that CagA uses a different domain to bind to PRK2 than to interact with Par1b/MARK2, we also tested modified CagA (CagA-FLP) that lost the ability to bind and inhibit Par1b/MARK2 kinase. As expected, CagA-FLP still inhibited PRK2 kinase activity further supporting the notion that PRK2 binds to a different CagA domain than Par1b/MARK2 (Fig. 5A).

The kinase assay described earlier used a protein substrate that was analysed by immunoblotting followed by densitometry for relative quantification of phosphorylated substrate band intensities. To further support our data, we also applied a different kinase assay that measured relative PRK2 kinase activity via determination of the conversion rate of adenosine triphosphate (ATP) into adenosine diphosphate (ADP) during the kinase reaction. This conversion is a direct result of the substrate phosphorylation reaction that requires the hydrolysis of ATP. The hydrolysis of ATP into ADP is proportional to the amount of substrate phosphorylation and can therefore serve as readout of kinase activity. The results for PRK activities obtained using this different assay format (Fig. 5B) confirmed the results described for densitometry (Fig. 5A). As an additional control, we used the PRK2 kinase inhibitor Y27632 that caused a similar inhibitory effect on PRK2 as purified CagA. In addition, PKLtide, which corresponds to a carboxyterminal PRK2 motif and that is known to cause PRK2 autoinhibition (Bauer *et al*., 2012), was also able to reduce PRK2 kinase activity to a similar extent as was observed for purified CagA.

Discussion

We and others previously established that CagA is injected into gastric host epithelial cells via a T4SS (Segal *et al*., 1999; Amieva *et al*., 2003; Bagnoli *et al*., 2005; Stein *et al*., 2013; Hatakeyama, 2014) and then interacts with various host molecules including SHP-2 (Higashi *et al*., 2002), ZO-1 (Amieva *et al*., 2003), Src kinase (Stein *et al*., 2002; Selbach *et al*., 2003; Tegtmeyer and Backert, 2011), Csk (Tsutsumi *et al*., 2003), E-cadherin (Murata-Kamiya *et al*., 2007; Kurashima *et al*., 2008; Oliveira *et al*., 2009), Grb-2 (Mimuro *et al*., 2002), c-met (Churin *et al*., 2003) and Crk (Suzuki *et al*., 2005) to modify host–cell signalling pathways that are associated with cell proliferation, cell motility or cytoskeletal rearrangements to name a few. Previously, we and Saadat *et al*. demonstrated that CagA binds and inhibits the serine–threonine kinase Par1b-MARK2, which causes disruption of apical–basolateral cell polarity in MDCK tissue culture cells (Saadat *et al*., 2007; Zeaiter *et al*., 2008).

In this study we show using AGS tissue culture infection assays that CagA causes the redistribution of the serine–threonine kinase PRK2 from the cytosol to the membrane, where it localized beneath the bacterial attachment site. Although PRK2 showed a similar CagA-dependent redistribution as Par-1b, both kinases appeared to be recruited independently from each other. Because Par-1b binding-deficient CagA still mediated PRK2 recruitment and inhibition, our results further suggest that (i) the CagA domain required for PRK2 binding does not involve the Par1b binding site (i.e. the TM motif) of CagA and (ii) PRK2 inhibition may not be mediated by binding of CagA to the catalytic active site of PRK2 as it is the case for Par1b. Although we currently cannot provide evidence for the latter, we suggest that inhibition of PRK2 is rather mediated by CagA binding to the potentially regulatory proline-rich motif of PRK2, which is located in the linker region between the C2-like domain of PRK2 and the catalytic kinase domain. However, PRK2 inhibition did not abolish PRK2 autophosphorylation in the kinase loop as the presence of CagA did not alter the total amount of phopho-PRK2 in the cells. Our hypothesis is supported by the finding that CagA interacts with PRK2, but not with the highly similar PRK1. Both kinases show a homology in the catalytic domains of 87%, with a homology within the N-terminal regulatory domains of only 48% (Bauer *et al*., 2012). Indeed, although the ACC domains as well as the C2-like domain are present in both kinases, the main difference between both protein kinases appears to be the presence of the proline-rich motif of PRK2, which constitutes a minimal SH3–protein interaction consensus motif (PxxPxRxxSL) (Ponting and Parker, 1996; Mukai, 2003). This motif, as mentioned previously, is absent in PRK1. In such a scenario, CagA-mediated PRK1 recruitment and inhibition would not be expected and could therefore explain the specificity of CagA towards PRK2 (Fig. 6). However, a SH3 domain has not been described for CagA. Experiments to identify the binding motifs in CagA and PRK2 and to confirm our hypothesis are currently under way in our laboratory. Although a functional role of the Pro-rich motif for PRK2 has not been established yet, previous manuscripts demonstrated interaction of this motif with the SH3 domain-containing protein Nck (Quilliam *et al*., 1996) and Grb4 (Braverman and Quilliam, 1999), two molecules implicated in cell morphological changes and transformation.

Small G-protein effectors of the Rho family include Rho and Rac and are well characterized as modulators of the actin cytoskeleton affecting cell adhesion, growth, cell morphology and cell motility (Vega and Ridley, 2008). Through these activities Rho proteins are thought to participate in cell invasion and metastasis, which are crucial events for cancer development. Because all PRK isoforms are activated by Rho GTPases (Mukai, 2003), it is not surprising that they have somewhat similar albeit not necessarily redundant functions affecting actin cytoskeletal dynamics. PRK1 has been described to trigger effects on the cytoskeleton by phosphorylating proteins such as α-actinin, CP-17 and intermediate filament proteins such as vimentin (Matsuzawa *et al*., 1997). PRK1 was also shown to be activated downstream of the androgen growth factor receptor and was implicated in cancer progression and is overexpressed in prostate cancer (Metzger *et al*., 2003; 2008). PRK3, which in humans appears restricted to skeletal muscle, heart and liver (Palmer *et al*., 1995), is activated via insulin and the PI3 kinase pathway leading to phosphorylation of the microtubule regulator CLIP-170 and the epidermal growth factor receptor (Leenders *et al*., 2004; Collazos *et al*., 2011). PRK3 is up-regulated in various tumour cell lines and following activation by RhoC-induced cell invasion and an advanced malignant pheno-type in breast cancer cells *in vitro* (Leenders *et al*., 2004). Similar to PRK1 and PRK3, only a few targets have been described for PRK2. PRK2 was activated by platelet derived growth factor receptor (PDGF) and CD44 receptor signalling and triggered various cellular responses including the phosphorylation and activation of the Src family kinase Fyn (Bourguignon *et al*., 2004; 2007; Boughan *et al*., 2006). Fyn in turn stabilized cell-to-cell adhesion by phosphorylation of β-catenin, which promotes β-catenin interactions with E-cadherin. CagA was shown to interact with E-cadherin and reduce E-cadherin/β-catenin interaction, thus leading to translocation of β-catenin to the nucleus. It is conceivable that inhibition of PRK2 by CagA could potentially reduce Fyn activation and thus further weaken E-cadherin/β-catenin interaction. Further supporting a role of PRK2 in cell-to-cell interactions is the recent finding that PRK2 promotes maturation of apical junctions downstream of Rho (Wallace *et al*., 2011). The motif within PRK2 that was required for recruitment to junctions and junctional maturation was identified as the C2-like domain. Therefore, CagA may exploit two serine–threonine kinases, PRK2 and Par1b, to disrupt cellular polarity.

Using transwell migration, scratch wound assays and Matrigel invasion chambers, Lachmann *et al*. (2011) demonstrated that active PRK2 kinase has a critical role in cell migration and invasion of the bladder tumour cell line 5673. This activity was dependent on the PRK2 regulatory domain, although no specific motifs were identified. The authors also observed that cells lacking PRK2 formed larger focal adhesions that might promote stronger adhesion and slow down cell migration. The fact that CagA was described to increase cell motility and cell invasive behaviour would therefore suggest that CagA activates PRK2 rather than inhibiting it. Our opposite findings may suggest that CagA-mediated stimulation of cell migration depends on SHP-2 binding and other factors, rather than on CagA/PRK2 interactions and that the latter are rather affecting junctional integrity. PRK2 has furthermore been described to phosphorylate cortactin, which was shown to reduce its F-actin cross-linking activity via the Arp2/3 complex (Bourguignon *et al*., 2004). Indeed, others have shown that kinase-dead PRK2 caused disruption of actin stress fibres in NIH3T3 cells (Vincent and Settleman, 1997).

Finally, PRK2 appears to be a target of other pathogens as well. HCV virus replication depended on PRK2-mediated phosphorylation of HCV RNA-dependent RNA polymerase (Kim *et al*., 2012; Han *et al*., 2014). Interaction between the YopM effector of *Yersinia* species and PRK2 was required to establish full virulence in animal models (McPhee *et al*., 2010; Hofling *et al*., 2014). Therefore, *H. pylori* is only the third pathogen described to interfere with PRK2 signalling.

In summary, our results demonstrate that CagA binds to and inhibits PRK2 kinase activity. We suggest that PRK2 inhibition may increase CagA-mediated effects on the host cell, possibly ranging from disruption of rho-mediated cytoskeletal rearrangements to destabilization of cellular junctions and/or cell adhesion. Our results also strengthen a model, where CagA has evolved as a master regulator of host kinases and phosphatases that are crucial for many host-signalling pathways. Although CagA interaction with SHP-2, c-Src family kinases, Csk and Par1b/MARK2 has been described previously, we here demonstrated that the serine–threonine kinase PRK2 may be another important target of the multifunctional toxin.

Experimental procedures

Bacterial strains and growth conditions

H. pylori strain G27 and its isogenic mutants, Δ*cag*A and Δrep^{EPISA}, were described previously (Stein *et al*., 2000). The strains ΔAxA and ΔAxA^{ΔFLP} were generated as follows. The *cag*A gene of strain 26695 (ATCC 700392) was amplified and cloned into pFLAG-CMV4 using the NotI/BamHI restriction sites. To create the Par1b binding-deficient strain (ΔAxA^{ΔFLP}), point mutations were introduced into *cag*A by polymerase chain reaction resulting in the following changes within CagA: L950G, R952G, K955A, L959A, K989A and L993A (CagA^{ΔFLP}) (vector kindly provided by Dr. Nesic, Rockefeller). Wild-type *cag*A and *cag*A^{ΔFLP} were then subcloned into a recombination that permitted homologous recombination of the *cag*A constructs into the recombinase gene of the G27 *cag*A mutant. Selection for recombinants was performed using the selection marker kanamycin, and all constructs were confirmed by DNA sequencing. All strains of *H. pylori* were cultured on Columbia agar plates supplemented with selective antibiotics and 5% horse blood and incubated at 37°C for 48 h using anaerobic jars containing Campygen (Oxoid) gas packs (10% CO_2, 5% O_2 and 85% N_2). Liquid cultures were made in Brucella broth supplemented with 10% fetal bovine serum (FBS) under aforementioned microaerophilic conditions. Inoculates were incubated overnight with shaking (165 r.p.m) at 37°C.

Immunofluorescence and confocal microscopy

AGS cells were seeded on sterile 12 mm glass coverslips inserted in a 24-well plate at a density of 3×10^5 cells ml^{-1} using Roswell Park Memorial Institute (RPMI) medium. After 18 h confluent monolayers were washed three times with infection medium (RPMI plus 5% FBS, pH 6.5). Infections with *H. pylori* strains were performed at a multiplicity of infection of 100:1 bacteria per cell for all assays except immunofluorescence microscopy experiments, where an infection ratio of 10:1 was used. Following infection, cells were washed briefly in wash buffer (TBS plus 1.5% FBS) and fixed with 4% (v/v) paraformaldehyde in TBS for 1 h. Then cells were washed and treated with 0.2% saponin in TBS for 10 min at room temperature. Coverslips were treated with primary antibodies diluted in wash buffer for 1 h at room temperature, followed by washing with TBS and treating with Alexa-Fluor conjugated secondary antibodies for 1 h at room temperature. Coverslips were mounted on glass slides using ProLong® Gold antifade reagent (Invitrogen). Slides were examined using a Leica DMI6000 B-inverted fluorescence microscope (Leica Microsystems), and fluorescent intensity was detected with an L5 or TX2 filter. Images were processed in OpenLab 5.0.2. For confocal microscopy, slides were analysed with a Zeiss LSM 510 inverted microscope and images were processed using ZeissZen 2009 lite software.

Partial purification of CagA proteins from* H. pylori *strains—*H. pylori* strains were cultured in *Brucella* broth with 10% FBS and incubated under microaerophilic condition at 37°C overnight with shaking. The incubated cultured broth was centrifuged at 4°C to pellet bacteria. The bacterial pellet was resuspended in ice-cold phosphate buffered saline (PBS) without calcium and magnesium (Sigma Life Science) in the presence of 1 mM phenylmethanesulphonylfluoride (PMSF; Sigma) and protease inhibitor cocktail (Sigma).

Sonication of the bacterial suspension was carried out on ice at 30% duty cycle with an output control of 3 for 15 s intervals (Sonicator, Ultrasonics, Inc.). The sonicated bacterial samples were cleared by repeated centrifugation at 14 000 r.p.m. at 4 °C. The clear supernatant samples were loaded to a Poly-Prep chromatography column that was filled with 1 ml of CHT™ ceramic hydroxylapatite type-1 (80 µM) (Bio-Rad), and pre-equilibrated in 20 mM sodium phosphate buffer, pH 7. Samples were allowed to pass through the CHT column under gravity followed by washing with two to three column volumes of 20 mM sodium phosphate buffer, pH 7, then with five column volumes of 300 mM sodium phosphate buffer, pH 7. Finally, the samples were eluted from column with 5 mL of 500 mM sodium phosphate buffer, pH 7, and collected in five 1 ml fractions. The eluted fractions were dialysed against PBS overnight using a Tube-0-Dialyzer 50 kDa Medi (G Bioscience) and then tested for purity using spectrophotometric analysis, sodium dodecyl sulfatepolyacrylamide gel electrophoresis (SDS-PAGE) and immunoblot analysis with anti-CagA antibodies. Samples were analysed for total protein concentration and preserved at −80°C for a very short period of time before they were used in immunoprecipitation and enzyme kinase assay experiments.

Cell fractionation

AGS cells were grown and infected as described earlier. Approximately 9×10^6 AGS cells were collected from culture dishes by scraping, pelleted and resuspended in 100 µl of ice-cold saponin buffer [50 mM Tris/Cl, pH 7.5, 1 mM PMSF, 1 mM Na_3VO_4, 1 mM NaF, saponin 1% (w/v), plus cocktail of protease inhibitors (complete, ethylenediaminetetraacetic acid-free, Roche)]. After 10 min on ice, the cells were pelleted (16 100× g for 5 min) and the supernatant transferred to a new tube (cytosol fractions). The pelleted cells were resuspended in 800 µl of saponin buffer and pelleted again. The supernatant was discarded and 100 µl of lysis buffer [saponin buffer plus 1% Nonidet P40 (v/v)] was added to solubilize the cell pellet on ice for 10 min. Lysates were pelleted to remove the Triton Nonidet P40 insoluble fraction and supernatants were transferred to new tubes (membrane fraction). Twenty-five microlitres of 5× Laemmli sample buffer (5% β-mercaptoethanol) was added to each of the cytosol and membrane fractions and samples were boiled for 10 min. For total cell lysates, 9×10^6 AGS cells were resuspended in 400 µl of PBS. One hundred microlitres of 5× Laemmli sample buffer was added, and cells lysates were boiled for 10 min.

Immunoprecipitation and immunoblotting—Briefly, the partially purified CagA was incubated on ice together with purified PKN2/PRK2 (recombinant active GST-tagged kinase – human PRECISIO^R kinase from Sigma) in immunoprecipitation buffer [TBS plus 1 mM PMSF and 1× protease inhibitor cocktail (Sigma)]. The mixture was then treated overnight with anti-PRK2 antibodies (Cell Signaling) in a rocking Eppendorf tube holder at 4°C, followed by addition of Dynabeads Protein G (Novex by Life Technologies) for 4 h. Prior to addition of Dynabeads, the beads were blocked with 2% BSA for 30 min to reduce non-specific protein binding. The next morning the beads were pelleted with a magnetic tube stand, and washed three times with immunoprecipitation buffer. Finally, protein samples for Western blot were prepared by eluting proteins from the Dynabeads by boiling for 2 min in 3× SDS protein sample buffer. The eluted protein samples were analysed by SDS-PAGE, followed by transfer onto a polyvinylidene difluoride membrane. The membranes were initially probed with anti-CagA primary antibody (b-300, Santa Cruz Biotechnology, Inc.), and were then treated with secondary antibodies conjugated to horseradish peroxidase. The development of blots was performed with SuperSignal^{R} West Pico chemiluminescent substrate (Thermo Scientific) and using a ChemiDoc™ imaging system (Bio-Rad). Later, the same blots were stripped with Restore™ Western Blot Stripping buffer (Thermo Scientific) and probed with anti-PRK2 primary antibodies (Cell Signaling Technology).

Enzyme kinase assay—The *in vitro* PRK2 enzyme kinase assays were performed in the presence of purified CagA, or alternatively mock-purified protein obtained from the isogenic *cag*A mutant strain. As a negative control, the kinase assay was also carried out using 1 μg BSA instead of purified CagA. The peptide substrate for PRK2, KKGRPRTSSFAEG (KKCrosstide), was derived from the GSK3 sequence, and a PRK2 inhibitor peptide PKLQRQKKIFSKQQG (PKLtide) (Bauer *et al*., 2012) was procured from Biomatik. The PRK2 kinase reaction was performed with PKC kinase assay kits (ImmunoChem) as per kit protocol, and the substrate phosphorylation levels were determined by measuring the detected band densities for phosphorylated substrate in Western blot using ChemiDoc MP imaging system program (Bio-Rad). However, for synthesized peptide substrate, ADP-Glo™ kinase assay kit (Promega) was used to estimate the PRK2 kinase activity by measuring the luminescence of the released in Infinite M200 (Tecan) as per manufacturer's instructions. In addition to PKLtide as a PRK2 kinase inhibitor, we also tested one chemical inhibitor Y27632 dihydrochloride (Sigma), which has been previously reported as an inhibitor for ROCK and PRK2 kinase (Shao *et al*. 2008). All data obtained from the experiments were analysed for comparison in MS Excel for mean ± standard deviation, and the standard error was determined using GraphPad software.

Cell culture and iTRAQ mass spectrometry analysis of phospho-proteins

The AGS human gastric adenocarcinoma cell line was obtained from ATCC (ATCC# CRL-1739) and cultured in RPMI 1640 medium (Invitrogen) supplemented with 10% FBS (Invitrogen). Cell lines were maintained at 37°C in 5% CO_2 atmosphere and 95% humidity. For ITRAQ analysis, 10 cm culture dishes (Nalgen) were seeded with 5×10^6 AGS cells in RPMI medium containing 10% FBS and incubated overnight at 37°C under 5% CO_2. The next day the AGS cells were washed with PBS containing magnesium and calcium (PBS++) and 5 ml fresh culture medium was added. Cells were infected with either *H. pylori* strain G27 wild type or the isogenic Δ*cag*A mutant (two dishes each). After 4 h, cells were washed again with PBS++, harvested by scraping and fractionated on ice into cytosol (1% saponin soluble fraction), membrane (1% Nonidet P40 soluble fraction) and unsoluble fractions as previously described (Zeaiter *et al*., 2008). The membrane fraction was then used for the isolation of phosphoproteins using a phosphoprotein isolation kit (Qiagen) following the manufacturer's instructions. The eluted phosphoproteins were then submitted for iTRAQ analysis as previously described (Zeaiter *et al*., 2008).

siRNA transfection—A total of 3×10^5 AGS cells were seeded in 5 cm tissue culture dishes over night to reach a confluency of approximately 60%. The next day the RPMI medium containing 10% FBS was replaced by fresh medium and the cells were transfected simultaneously with two PRK2 siRNA(h) (Ambion, cat. # AM51333; ID: 780 AND 781) or alternatively with control siRNA (Ambion, cat. # AM4611). Transfection of siRNA was performed using Lipofectamine® RNAiMAX (Life Technologies, cat. # 13778030) following the manufacturer's recommendations. Forty hours after transfection, the medium was exchanged for fresh medium again and cells were infected with *H. pylori* and fractionated as described earlier.

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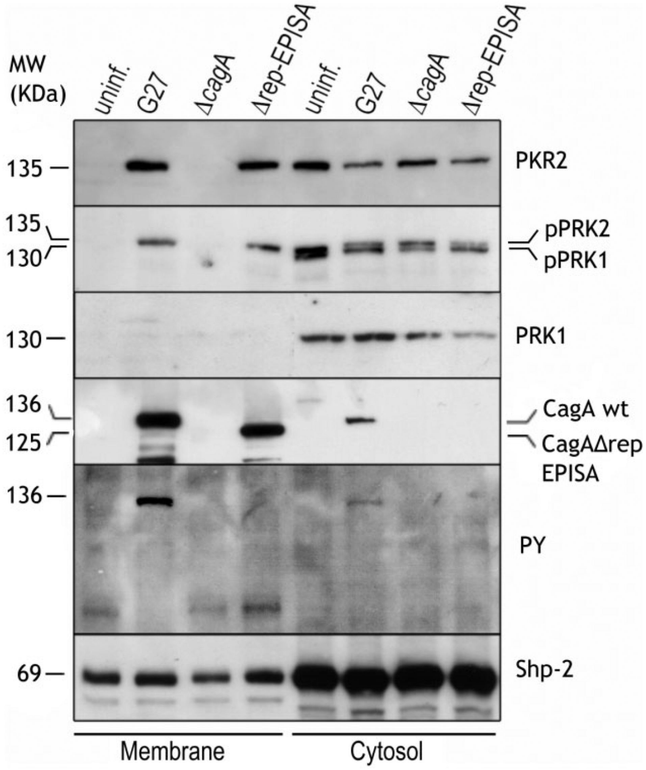

Fig. 1.

CagA causes redistribution of PRK2 from the cytosol to the membrane. AGS cells were infected with *H. pylori* strains as indicated. After 4 h of infections, cells were collected and fractionated into cytosol and membrane fractions, which were analysed by immunoblotting with various antibodies as shown.

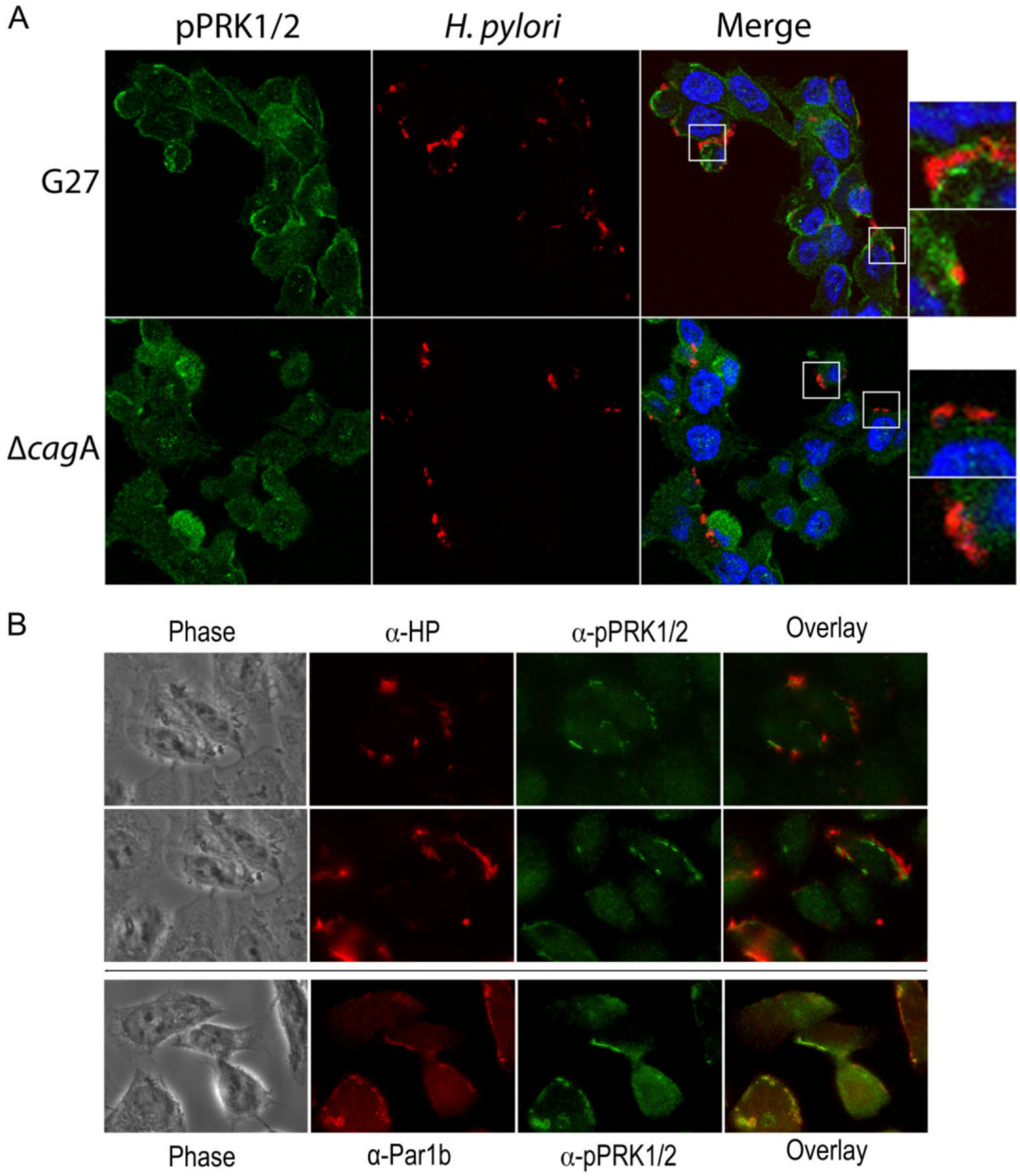

Fig. 2.

PRK2 is recruited by *H. pylori* dependent on CagA.

A. AGS cells were infected with *H. pylori* wild-type strain G27 or the isogenic *cag*A mutant (Δ*cag*A). After 3 h, cells were fixed with 4% paraformaldehyde and permeabilized with 0.2% saponin. Cells were stained with anti-*H. pylori* antibody, anti-phospho-PRK1/2 antibody and 4′,6-diamidino-2-phenylindole (DAPI), and analysed by confocal microscopy (63× magnification). White boxes indicate areas of additional magnification.

B. G27 infected AGS cells were processed as above and treated with a combination of either anti-*H. pylori* (anti-HP) and pPRK1/2 antibodies (upper panel) or a combination of anti-Par1b and pPRK1/2 antibody (lower panel) and analysed by wide field and fluorescence microscopy (100× magnification). Phase contrast pictures were added to show the morphology and cell borders of AGS cells.

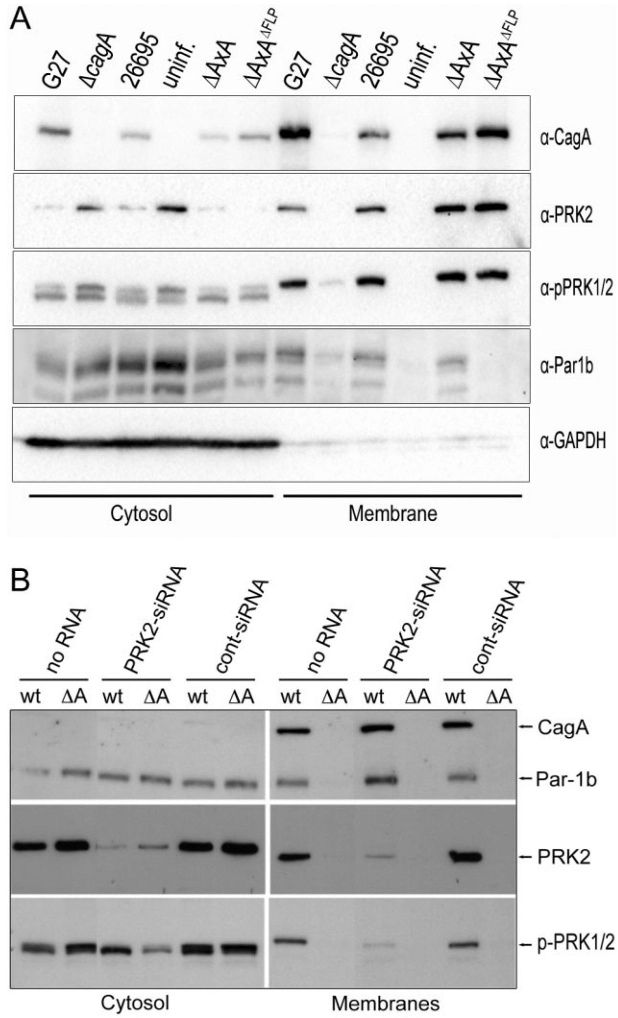

Fig. 3.

CagA recruits PRK2 and Par1b independently from each other.

A. AGS cells were infected with *H. pylori* strains as indicated. After 3 h, cells were collected and fractionated into cytosol and membrane fractions, which were analysed by immunoblotting with various antibodies as shown.

B. AGS cells were treated with siRNA against PRK2, or control siRNA, or left untreated. After 48 h, cells were infected with either *H. pylori* stain G27 (Wt) or the isogenic Δ*cag*A strain for 3 h and then fractionated into cytosol and membrane. The fractions were analysed by immunoblotting using various antibodies as indicated.

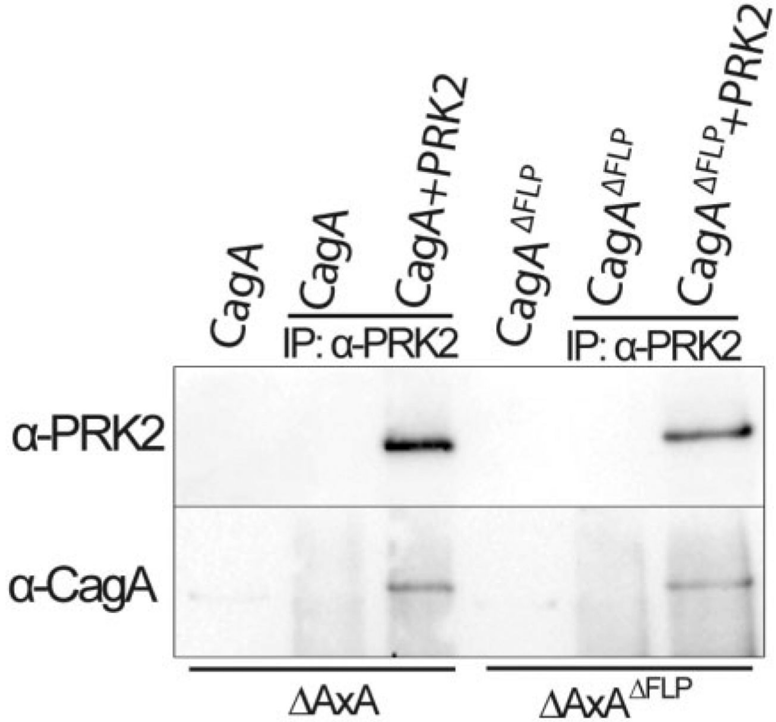

Fig. 4.

CagA co-immunoprecipitates with PRK2. CagA was partially purified from *H. pylori* strains ΔAxA or, alternatively, ΔAxA$^{\Delta FLP}$ using ceramic hydroxylapatite (CHT) resin. The partially purified proteins were then used in the presence or absence of purified PRK2 for co-immunoprecipitation studies. Purified CagA and immunoprecipitates were then analysed by immunoblotting using anti-CagA or anti-PRK2 antibodies.

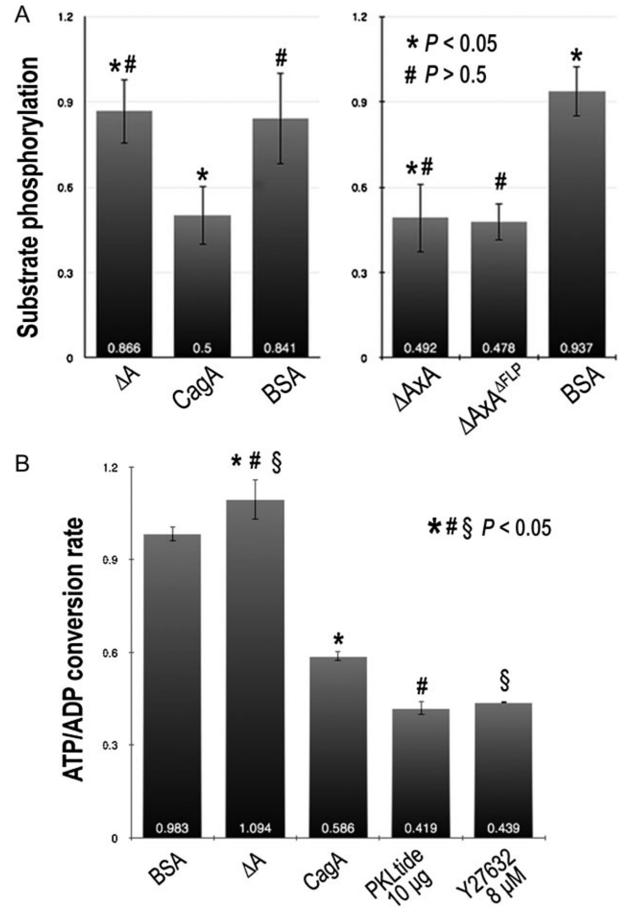

Fig. 5.

Effect of CagA on *in vitro* enzymatic activities of PRK2 kinase. CagA was partially purified from *H. pylori* strains G27 (CagA), Δ*cag*A (ΔA), ΔAxA or alternatively $\Delta AxA^{\Delta FLP}$ and used together with purified PRK2 kinase in *in vitro* kinase assays. BSA was used as a negative control. The two-tailed *P*-values were calculated in *t*-test using GraphPad software. The *t*-test results indicate differences in kinase activity that were statistically significant ($P < 0.05$) or not statistically significant ($P > 0.5$).

A. A PKC kinase assay kit was used to determine PRK2 kinase activity. Relative quantification of substrate phosphorylation was performed by immunoblotting of kinase substrate and determination of band intensities using densitometry.

B. The ADP-Glo kinase assay kit (Promega) was used to estimate the PRK2 kinase activity by measuring the luminescence generated during the kinase reaction by conversion of ATP into ADP.

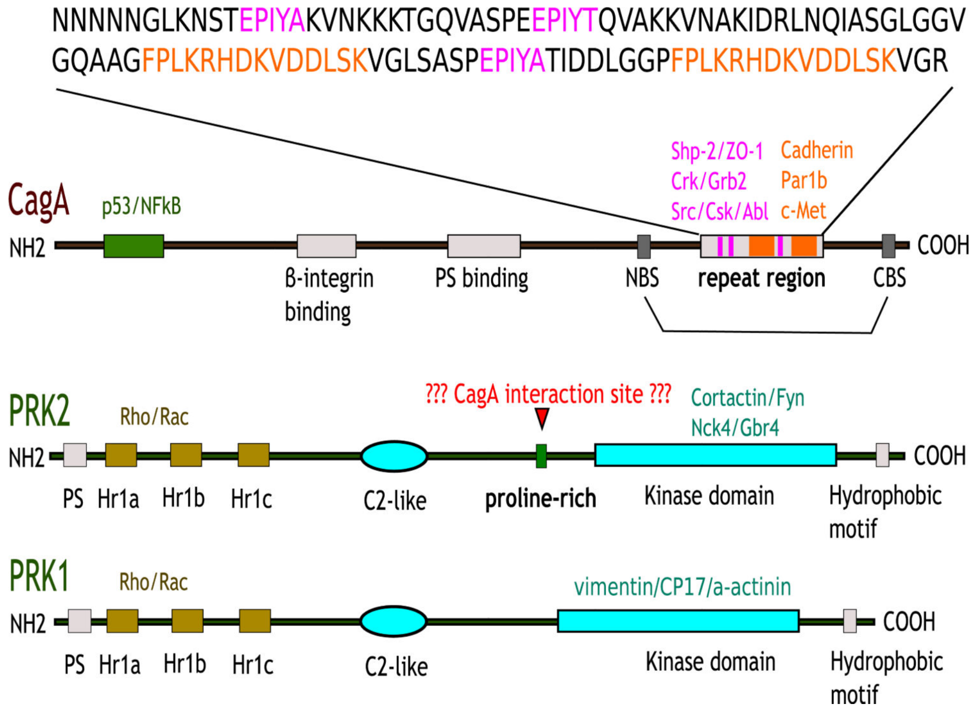

Fig. 6.

Schematic model of CagA and PRK1/PRK2 functional domains. CagA: The amino acid sequence of the CagA (strain 26695) repeat region containing the EPIYA/T motifs (purple) and the CagA dimerization motifs (TM motifs; orange) are shown. Interacting signalling molecules are indicated above the respective proteins, and structural domains below. CBS, C-terminal binding sequence; NBS, N-terminal binding sequence.